ANNEX 14

RESOLUTION MEPC.376(80)

(adopted on 7 July 2023)

GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS

(LCA GUIDELINES)

THE MARINE ENVIRONMENT PROTECTION COMMITTEE,

RECALLING Article 38(a) of the Convention on the International Maritime Organization (the Committee) concerning the functions of the Marine Environment Protection Committee conferred upon it by international conventions for the prevention and control of marine pollution from ships,

RECALLING ALSO that at its seventy -second session, the Committee adopted resolution MEPC.304(72) on Initial IMO Strategy on Reduction of GHG Emissions from Ships (Initial Strategy),

NOTING that the Initial Strategy calls for the development of robust life cycle GHG/carbon intensity guidelines for all types of fuels, in order to prepare for an implementation programme for effective uptake of alternative low-carbon and zero-carbon fuels,

NOTING ALSO that at its eightieth session, the Committee adopted resolution MEPC.377(80)

for effective uptake of alternative low-carbon and zero-carbon fuels,
NOTING ALSO that at its eightieth session, the Committee adopted resolution MEPC.377(80) on the 2023 IMO Strategy on Reduction of GHG Emissions from Ships (2023 IMO Strategy) setting out the levels of ambition for the international shipping sector in reducing GHG emissions,
NOTING FURTHER that the 2023 IMO Strategy provides that the levels of ambition and indicative checkpoints should take into account the well -to-wake GHG emissions of marine fuels as addressed in the guidelines on life cycle GHG intensity of marine fuels dev eloped by the Organization,
NOTING that the 2023 IMO Strategy provides that the basket of candidate mid -term GHG reduction measures should take into account the well-to-wake GHG emissions of marine fuels as addressed in the guidelines on life cycle GHG intensity of marine fuels developed by the Organization,
HAVING CONSIDERED, at its eightieth session, the draft guidelines on life cycle GHG intensity of marine fuels (LCA Guidelines),
1 ADOPTS the Guidelines on life cycle GHG intensity of marine fuels (LCA Guidelines), as set out in the annex to the present resolution;

intensity of marine fuels (LCA Guidelines),
1 ADOPTS the Guidelines on life cycle GHG intensity of marine fuels (LCA Guidelines),
as set out in the annex to the present resolution;
2 AGREES that any regulatory application and implications of the LCA Guidelines
should be determined by the Committee in the process of developing regulatory provisions,
3 REQUESTS Member Governments to bring the annexed Guidelines to the attention
of shipowners, ship operators, shipbuilders, ship designers, energy companies, fuel producers,
bunkering companies, engine manufacturers and any other interested parties;
4 AGREES to keep these Guidelines under review in light of experience gained with
their implementation.
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
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GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS

(LCA Guidelines)

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PART I: GENERAL

1 INTRODUCTION

1.1 These guidelines provide guidance on life cycle GHG intensity assessment for all fuels

PART I: GENERAL

1 INTRODUCTION

1.1 These guidelines provide guidance on life cycle GHG intensity assessment for all fuels and other energy carriers (e.g. electricity) used on board a ship. These guidelines aim at covering the whole fuel life cycle (with specific boundaries), from feedstock extraction/cultivation/ recovery, feedstock conversion to a fuel product, transportation as well as distribution/bunkering, and fuel utiliza tion on board a ship. These guidelines also specify sustainability themes/aspects for marine fuels and define a Fuel Lifecycle Label (FLL), which carries information about fuel type, feedstock (feedstock type and feedstock nature/carbon source), conversion/production process (process type and energy used in the process), GHG emission factors, information on fuel blends and sustainability themes/aspects.
These guidelines specify the elements of FLL subject to verification/certification and include a general procedure on how the certification scheme/standards could be identified.

2 SCOPE

2.1 The scope of these guidelines is to address well-to-tank (WtT), tank-to wake (TtW),

general procedure on how the certification scheme/standards could be identified.

2 SCOPE

2.1 The scope of these guidelines is to address well-to-tank (WtT), tank-to wake (TtW), and well-to-wake (WtW) greenhouse gases (GHG) intensity and sustainability themes/aspects related to marine fuels/energy carriers (e.g. electricity for shore power) used for ship propulsion and power generation onboard. The relevant GHGs included are carbon dioxide (CO 2), methane (CH4) and nitrous oxide (N2O). These guidelines are not intended to provide guidance for a complete IMO GHG inventory for international shipping. Emissions from cargo (e.g. volatile organic compounds (VOC)), or use of refrigerants are not included; other short -lived climate forcers and precursors such as non -methane volatile organic compounds (NMVOC), sulphur oxides (SO x), carbon monoxide (CO), particulate matter (PM) and Black Carbon are not part of the scope of these LCA guidelines.

2.2 The system boundaries of the WtW GHG emission factors calculation, in the context of these guidelines span the life cycle of fuels from their sourcing to production, conversion,

not part of the scope of these LCA guidelines.

2.2 The system boundaries of the WtW GHG emission factors calculation, in the context of these guidelines span the life cycle of fuels from their sourcing to production, conversion, transport, distribution, and eventually their use on board ships based on an attributional approach.1 The possibility to expand the system boundaries for specific pathways in which the feedstock is displaced from present use(s) will be assessed on a case-by-case basis.2 As such, emissions associated with the following life cycle stages of the fuel life cycle chain will be accounted for:

.1 feedstock extraction/cultivation/acquisition/recovery;
.2 feedstock (early) processing/ transformation at source;
.3 feedstock transport to conversion site;
.4 feedstock conversion to product fuel;
.5 product fuel transport/storage/delivery/retail storage/bunkering; and
.6 fuel utilization on board a ship.

2.3 Consistently with the attributional approach and using best available scientific evidence, the WtT emissions calculations (i.e. emissions related to the fuel sourcing,

.6 fuel utilization on board a ship.

2.3 Consistently with the attributional approach and using best available scientific evidence, the WtT emissions calculations (i.e. emissions related to the fuel sourcing, production, conversion, transport and delivery) are assessed regardless of the final use of

1 Attributional Life Cycle Assessment (LCA): LCA aiming to describe the environmentally relevant physical flows to and from a system and its subsystems over their life cycle; Consequential Life cycle Analysis (LCA): LCA aiming to describe how environmentally relevant flows will change in response to poss ible decisions. (Finnveden G, Hauschild MZ, Ekvall T, Guinée J, Heijungs R, Hellweg S, et al. "Recent developments in life cycle assessment". Journal of Environmental Management. 2009;91(1):1-21).

2 Such as for captured CO2 transportation and storage.

fuels/energy carriers, and the TtW emissions (i.e. emissions related to the fuel use) are

fuels/energy carriers, and the TtW emissions (i.e. emissions related to the fuel use) are quantified regardless of the sourcing/production/conversion/transport and delivery steps of the fuel/energy carrier. WtW emissions are given by the sum of the two par ts, providing the full emission performance associated with the fuel production and use of a certain fuel/energy in a specific converter onboard.

2.4 The GHG emissions are calculated as CO 2-equivalent (CO 2eq), using the Global Warming Potential over a 100 -year time -horizon (GWP100) to convert emissions of other gases than CO 2, as given in the fifth IPCC Assessment Report, 3 for CO 2, CH 4 and N2O, as follows:

$$2(100) = 2(100) \times 2 + 4(100) \times 4 + 2(100) \times 2$$

(CO2 1; CH4 28; N2O 265), this would read as:

$$2(100) = 1 \times 2 + 28 \times 4 + 265 \times 2)$$

These GWP100 values should be used for the purpose of quantifying the GHG intensity in accordance with these guidelines.

2

(CO2 1; CH4 28; N2O 265), this would read as:

2(100) = 1 × 2 + 28 × 4 + 265 × 2)

These GWP100 values should be used for the purpose of quantifying the GHG intensity in accordance with these guidelines.

A calculation using a Global Warming Potential over a 20 -year horizon (GWP20) may be provided as information for comparative purposes, as follows:

2(20) = 2(20) × 2 + 4(20) × 4 + 2(20) × 2

(CO2 1; CH4 84; N2O 264), this would read as:

2(20) = 1 × 2 + 84 × 4 + 264 × 2

2.5 These guidelines provide:

.1 WtW GHG emission factors based on a life cycle attributional methodology, expressing the GHG profile of each representative fuel using on Global Warming Potential (GWP) values over a 100 -year time-horizon of included GHG (CO2, CH4 and N2O);

.2 WtT GHG emission factors (CO2, CH4 and N2O) quantified consistently with the attributional approach;

.3 TtW GHG emission factors (CO2, CH4 and N2O); and

.4 sustainability themes/aspects for marine fuels.

2.6 These guidelines define a FLL that carries information about fuel type, feedstock

the attributional approach;

.3 TtW GHG emission factors (CO2, CH4 and N2O); and

.4 sustainability themes/aspects for marine fuels.

2.6 These guidelines define a FLL that carries information about fuel type, feedstock used, fuel production pathway, GHG emission factors, information on fuel blends and sustainability themes/aspects.

3 The Global Warming Potential values as given in the IPCC Fifth Assessment Report (AR5) are used in the context of these guidelines.

2.7 The figure below shows a generic WtW supply chain for a fuel. The bunkering marks the last step in the WtT phase before the TtW phase starts.

Figure 1: Generic well-to-wake supply chain

2.8 These guidelines include an initial non-exhaustive list of fuels in appendix 1 depicting the main current and expected future marine fuels.

PART II: METHODOLOGY

3 GENERAL APPROACH

3.1 A Life Cycle Assessment (LCA) based approach provides a holistic assessment of

the main current and expected future marine fuels.

PART II: METHODOLOGY

3 GENERAL APPROACH

3.1 A Life Cycle Assessment (LCA) based approach provides a holistic assessment of the product/service/system from well -to-wake using data specific to the activity considered. The LCA methodology follows the marine fuel from feedstock sourcing to its utilizati on onboard ships and assesses its life cycle GHG intensity. This approach, applied within the boundaries of the WtW GHG emissions quantification, is applicable across all geographical regions, where emissions occur and allows for quantifying the GHG intensity over the entire fuel/energy supply chain.

3.2 General principles and methodology can be found in ISO 14044:2006 Environmental management Lifecycle assessment Requirements and guidelines . ISO 14040:2006 Environmental management Lifecycle assessment Principles and framework sets the framework for the LCA, for the quantification of the environmental impact of products, processes and services in the supply chain. On this basis, a specific LCA methodology can be tailored for its application to marine fuels.

framework for the LCA, for the quantification of the environmental impact of products, processes and services in the supply chain. On this basis, a specific LCA methodology can be tailored for its application to marine fuels.

3.3 WtT emissions represent GHG emissions resulting from growing or extracting raw materials, producing and transporting the fuel up to the point of use, including bunkering.

3.4 TtW emissions represent GHG emissions resulting from fuel utilization onboard (e.g. combustion), including potential leaks (fugitive emissions and slip), when relevant for the GHG assessment.

3.5 WtW emissions are the sum of the WtT and TtW emissions and quantify the full life cycle GHG emissions for a given fuel and fuel pathway, used in a given energy converter on board.

3.6 The attributional approach considers all processes along the supply chain of fuel/energy carrier pathways, allowing the quantification of contributions per segment to the overall GHG intensity of the final fuel/energy product used on board a ship. The expansion of the system boundaries for specific pathways, in which the feedstock or intermediate products

overall GHG intensity of the final fuel/energy product used on board a ship. The expansion of the system boundaries for specific pathways, in which the feedstock or intermediate products are diverted from existing use(s), may be considered on a case-by-case basis.

3.7 As regards the expansion of the system boundaries, with consequential elements such as Indirect Land Usage Change (ILUC), concerns with respect to uncertainties and the risk of arbitrariness suggest that the feedstocks with associated ILUC should only be assessed through a risk-based approach, in the framework of sustainability themes/aspects, as part of these guidelines.

3.8 When more than one product results from a conversion process, emissions related to the fuel production should be allocated between main product and co -products. Within such conversion processes, emissions are allocated using their energy content, the so -called

the fuel production should be allocated between main product and co -products. Within such conversion processes, emissions are allocated using their energy content, the so -called "energy allocation" approach. Where co -products allocation cannot be performed based on their energy content (e.g. Oxygen resulting from water electrolysis for H 2 production), other methods such as mass allocation, market revenue allocation (also known as "economic allocation"), could be considered on a case-by-case basis.

3.9 A co-product is defined as "an outcome of a production process, which has economic value and elastic supply (intended as the existence of a clear evidence of the causal link between feedstock market value and the quantity of feedstock that can be produced)".

3.10 This definition applies also when a raw material used to produce the fuels is a waste (no economic value) or a residue (unavoidably produced and with negligible economic value, needing further processing to be used in the main conversion process). In case the feedstock is a waste, a residue or a by -product, emissions considered as WtT start at the feedstock

needing further processing to be used in the main conversion process). In case the feedstock is a waste, a residue or a by -product, emissions considered as WtT start at the feedstock collection point onwards until the point of use of the final fuel/energy product.

3.11 According to the IPCC Guidelines for National Greenhouse Gas Inventories ("the IPCC Guidelines"),4 any carbon in the fuel derived from biomass should be reported as an information item and not included in the sectoral or national totals to avoid double counting, since the net emissions from biomass are already accounted for in the Agriculture Forestry and Other Land Use (AFOLU) sector at a national level.

3.12 The scope of the IMO LCA guidelines does not affect or change the IPCC Guidelines. According to the IPCC Guidelines, international waterborne navigation (international bunkers) is grouped under "Mobile combustion" under the Energy sector, but emissions from fuel used by ships in international transport should not be included in national totals in national GHG inventories.

3.13 A fuel batch may be a mix of fuels made from various feedstocks and sources

by ships in international transport should not be included in national totals in national GHG inventories.

3.13 A fuel batch may be a mix of fuels made from various feedstocks and sources (e.g. by blending 20% biodiesel into fossil MGO) and/or through different production pathways. The calculation should be done using the weighted averages of the energy of the various fuel components. Relevant information should accompany each component fuel in the FLL. Blended fuels should be included in the certification schemes and relevant GHG default or actual emission factors (gCO2/MJ) determined in proportion to the energy of each fuel part of the blend.

4 WELL-TO-TANK (WtT)

4.1 The pathway of each relevant marine fuel should be clearly described and the GHG emissions during each step of the fuel pathway should be calculated. Specific GHG emissions of a specific non -conventional and non -fossil fuel's pathway may take into acco unt different characteristics across geographic regions, where feedstock production and/or conversion occurs, as appropriate.

4 2006 IPCC Guidelines for National Greenhouse Gas Inventories

characteristics across geographic regions, where feedstock production and/or conversion occurs, as appropriate.

4 2006 IPCC Guidelines for National Greenhouse Gas Inventories
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4.2 Any further reference in this document to a "fuel pathway" should be understood to include the feedstock structure (the so -called nature/carbon source and feedstock type pair) and the production or conversion process (noting that the same feedstock and fuel type pair can have a different production or conversion process).

4.3 The aim of the WtT methodology is to quantify and evaluate the GHG intensity of fuel production, including all steps mentioned in figure 2. The carbon feedstock and production pathway of a fuel should be identified in order to apply the methodology and is included as part of the FLL. The production steps to be included in the WtT are presented in figure 2.

Figure 2: Generic well-to-tank supply chain

pathway of a fuel should be identified in order to apply the methodology and is included as part of the FLL. The production steps to be included in the WtT are presented in figure 2.

Figure 2: Generic well-to-tank supply chain

4.4 The WtT GHG emission factor (gCO 2eq/MJ(LCV) fuel or electricity) is calculated according to Equation (1).

Equation (1)

 = + + +

Term	Units	Explanation
	gCO2eq / MJ(LCV)	Emissions associated with the feedstock extraction/cultivation/acquisition/recovery
	gCO2eq / MJ(LCV)	Emissions (annualized emissions (over 20 years) from carbon stock changes caused by direct land-use change)5
	gCO2eq / MJ(LCV)	Emissions associated with the feedstock processing and/or transformation at source and emissions associated with the conversion of the feedstock to the final fuel product, including electricity generation
	gCO2eq / MJ(LCV)	Emissions associated with the feedstock transport to conversion plant, and the emissions associated with the finished fuel transport and storage, local delivery, retail storage and bunkering

	gCO2eq / MJ(LCV)	Emissions associated with the feedstock transport to conversion plant, and the emissions associated with the finished fuel transport and storage, local delivery, retail storage and bunkering
	gCO2eq / MJ(LCV)	Emissions (annualized emission savings (over 20 years) from soil carbon accumulation via improved agricultural management)6

5 Pending further methodological guidance to be developed by the Organization, the value of parameter should be set to zero.

6 Pending further methodological guidance to be developed by the Organization, the value of parameter should be set to zero.

Term	Units	Explanation
	gCO2eq / MJ(LCV)	Emissions credit from carbon capture and storage (e ccs), that have not already been accounted for in e p. This should properly account the avoided emissions through the capture and sequestration of emitted CO 2, related to the extraction, transport, processing and distribution of fuel (c sc). From the

have not already been accounted for in e p. This should properly account the avoided emissions through the capture and sequestration of emitted CO 2, related to the extraction, transport, processing and distribution of fuel (c sc). From the above-mentioned emission credit, all the emissions resulting from the process of capturing (e cc) and transporting (e t) the CO2 up to the final storage (including the emissions related to the injection, etc.) need to be deducted.
This element should be calculated with the following formula:

=

g CO2 stored /
MJ(LCV)
Emissions credit equivalent to the net CO 2 captured and stored (long-term: 100 years)
gCO2eq /
MJ(LCV)
Emissions associated with the process of capturing, compression and/or cooling and temporary storage of the CO2
gCO2eq /
MJ(LCV)
Emissions associated with transport to a long -term storage site
gCO2eq /
MJ(LCV)
Any emissions associated with the process of storing (long -term: 100 years) the captured CO 2 (including fugitive emissions that may happen during long -term storage and/or the injection of CO2 into the storage)
gCO2eq /
MJ(LCV)

MJ(LCV)
Any emissions associated with the process of storing (long -
term: 100 years) the captured CO 2 (including fugitive
emissions that may happen during long -term storage and/or
the injection of CO2 into the storage)
gCO2eq /
MJ(LCV)
Any additional emissions related to the CCS

4.5 The WtT emissions in Equation (1) include emissions associated with raw materials extraction or cultivation, primary energy sources used for production of goods and utilities such as energy carriers (e.g. fuels and electricity), transport and distribution (including bunkering), direct land use change and changes in carbon stocks (soil carbon accumulation).

4.6 Processing incorporates all steps and operations needed for the extraction, capture or cultivation of the primary energy source. Process includes basic transformation at source and operations needed to make the resource transportable to the marketplace (e. g. drying, chemical/physical upgrade such as gas-to-liquid, etc.).

4.7 Transportation, processing and distribution include transportation of the products in the fuel pathway to the place of transformation, conditioning (such as compression, cooling),

chemical/physical upgrade such as gas-to-liquid, etc.).

4.7 Transportation, processing and distribution include transportation of the products in the fuel pathway to the place of transformation, conditioning (such as compression, cooling), distribution to the marketplace (i.e. bunkering) and eventual leakages, as well as fugitive emissions at any of these stages.

4.8 Allocation of emissions to co-products based on their energy content should be used, as the most appropriate and reliable methodology considering the establishment of an appropriate certification method using values that are predictable, reproducible and stable.

4.9 Land use (direct and indirect) for the production of biofuels may lead to land use change (LUC). LUC can be classified as direct LUC (DLUC) and indirect LUC (ILUC).

4.10 The DLUC definition is based on ISO 14067:2018 described as a change in the use or management of land within the product system being assessed. The DLUC impacts

4.10 The DLUC definition is based on ISO 14067:2018 described as a change in the use or management of land within the product system being assessed. The DLUC impacts comprises the emissions and sequestration resulting from carbon stock changes in biomass, dead organic matter and soil organic matters, evaluated in accordance with the IPCC Guidelines. When available, sector or country -specific data on carbon stocks may be used; otherwise, IPCC's Tier 1 default emission factors may be considered. Two terms in the W tT Equation (1) capture respectively emissions resulting from direct land use change, i.e. el, and sequestration or otherwise increase in the content of soil organic carbon: esca.

4.11 The ILUC definition is based on ISO 14067:2018, described as a change in the use or management of land, which is a consequence of direct land use change, but which occurs outside the product system being assessed. ILUC occurs as a result of the economic impacts induced by increased biofuel demand on commodity prices with resulting shifts in demand and

outside the product system being assessed. ILUC occurs as a result of the economic impacts induced by increased biofuel demand on commodity prices with resulting shifts in demand and supply across economic sectors, including primarily food and feed production. ILUC cannot be directly measured and is projected with economic models instead.

4.12 Owing to the variability of assumptions underlying the evaluation of indirect effects, quantitative assessment of GHG effects of ILUC is subject to uncertainty, high quantitative variability and to the risk of arbitrary conclusions. For these reasons, ILUC shou ld be at this stage addressed using a risk -based approach, meaning that quantitative values will not be calculated and assigned to each fuel pathway. The ILUC emissions, as well as the spatial dimension of the ILUC effects, are dependent on a var iety of factors such as local/regional conditions and practices for agriculture, current and expected food import demand, national current accounts, the type of feedstock, the alternative economic uses of the same feedstock, etc.

4.13 A qualitative risk-based approach to ILUC includes consideration on the following:

current accounts, the type of feedstock, the alternative economic uses of the same feedstock, etc.

4.13 A qualitative risk-based approach to ILUC includes consideration on the following:

.1 Low-ILUC risk qualifies and characterizes biofuel production projects that supply additional feedstock without disrupting existing land uses. When productivity is increased on an area which is in agricultural production, only additional yields should be considered as low-ILUC rather than the entire production; and

.2 High-ILUC risk qualifies and characterizes biofuel production projects based on, or displacing, food and feed crops resulting in a significant expansion of the feedstock production area shifting into land with high-carbon stock.

4.14 WtT default emission factors are provided in appendix 2 of these guidelines.

5 TANK-TO-WAKE (TtW)

5.1 The aim of the TtW methodology is to quantify and evaluate the intensity of CO2, CH4 and N2O emitted on board a ship related to the fuel usage, including combustion/conversion and all relevant fugitive emissions with a Global Warming Potential.

and N2O emitted on board a ship related to the fuel usage, including combustion/conversion
and all relevant fugitive emissions with a Global Warming Potential.

5.2 The TtW GHG emission factors should be calculated using Equation (2):

Equation (2)

= 1 ((1 1100 (_ + C)) × (2 × 2 + 4 × 4 + 2 × 2) + + (1100 (_ + C) × ×) × ×)

Note: terms , and are pending further methodological guidance to be developed by the Organization.
For more details refer to footnotes 11 to 13.

Term	Units	Explanation
Cslip_ship	% of total fuel mass	Factor accounting for fuel (expressed in % of total fuel mass delivered to the ship) which escapes from the energy converter without being oxidized (including fuel that escapes from combustion chamber/oxidation process and from crankcase, as appropriate)
_ = (1 /100)		
Cslip		

delivered to the ship) which escapes from the energy converter without being oxidized (including fuel that escapes from combustion chamber/oxidation process and from crankcase, as appropriate)

_ = (1 /100)
Cslip

% of total fuel mass
Factor accounting for fuel (expressed in % of total fuel mass consumed in the energy converter) which escapes from the energy converter without being oxidized (including fuel that escapes from combustion chamber/oxidation process and from crankcase, as appropriate)

Cfug % of fuel mass Factor accounting for the fuel (expressed in % of mass of the fuel delivered to the ship) which escapes between the tanks up to the energy converter which is leaked, vented or otherwise lost in the system7

Csfx

gGHG/g fuel Factor accounting for the share of GHG in the components of the fuel (expressed in g GHG/g fuel)
Example: for LNG this value is 1

CfCO2

gCO2/g fuel CO2 emission conversion factor (gCO 2/g fuel completely combusted) for emissions of the combustion and/or oxidation process of the fuel used by the ship

CfCH4

gCH4/g fuel CH4 emission conversion factor (gCH 4/g fuel delivered to

gCO2/g fuel CO2 emission conversion factor (gCO 2/g fuel completely combusted) for emissions of the combustion and/or oxidation process of the fuel used by the ship
CfCH4

gCH4/g fuel CH4 emission conversion factor (gCH 4/g fuel delivered to the ship) for emissions of the combustion and/or oxidation process of the fuel used by the ship8
CfN2O

gN2O/g fuel N2O emission conversion factor (gN 2O/g fuel delivered to the ship) for emissions of the combustion and/or oxidation process of the fuel used by the ship
GWPCH4

gCO2eq/g CH4 Global Warming Potential of CH4 over 100 years (based on the fifth IPCC Assessment Report 5)9
Definition as per https://www.ipcc.ch/assessment-report/ar5/

7 Pending further methodological guidance to be developed by the Organization to determine appropriate factor(s), the value of Cfug should be set to zero.

8 For LNG/CNG fuel, the Cslip_engine is covering the role of CfCH4, so CfCH4 is set to zero for these fuels.

9 Set at 28 based on IPCC AR5.

9 Set at 28 based on IPCC AR5.

Term Units Explanation
GWPN2O

gCO2eq/g N2O Global Warming Potential of N2O over 100 years (based on the fifth IPCC Assessment Report 5) .10 Definition as per https://www.ipcc.ch/assessment-report/ar5/

gCO2eq/g GHG Global Warming Potential of GHG in the components of the fuel over 100 years (based on the fifth IPCC scientific Assessment Report)

0 or 1 Carbon source factor to determine whether the emissions credits generated by biomass growth are accounted for in the calculation of the TtW value

gCO2eq/g fuel Emissions credits generated by biomass growth
11

gCO2eq/g fuel Emission credits from the used captured CO 2 as carbon stock to produce synthetic fuels in the fuel production process and utilization (that was not accounted under e fecu and ep)
12

0 or 1 Carbon source factor to determine whether the emissions credits from the used captured CO 2 as carbon stock to

process and utilization (that was not accounted under e fecu and ep)
12

0 or 1 Carbon source factor to determine whether the emissions credits from the used captured CO 2 as carbon stock to produce synthetic fuels in the fuel production process are accounted for in the calculation of the TtW value

13
gCO2eq / g fuel Emission credit from carbon capture and storage (e occs), where capture of CO2 occurs onboard. This should properly account for the emissions avoided through the capture and sequestration of emitted CO 2, if CCS occurs on board.
From the above -mentioned emission credit, all the emissions resulting from the process of capturing (ecc), and transporting (et) the CO 2 up to the final storage (including the emissions related to the injection, etc.) need to be deducted.
This element should be calculated with the following formula:

=

gCO2 / g fuel Credit equivalent to the CO 2 captured and stored (long - term: 100 years)
gCO2eq / g fuel Any emission associated with the process of capturing, compress and temporarily store on board the CO2

10 Set at 265 based on IPCC AR5.

term: 100 years)
gCO2eq / g fuel Any emission associated with the process of capturing, compress and temporarily store on board the CO2

10 Set at 265 based on IPCC AR5.

11 Pending further methodological guidance to be developed by the Organization, the value of the multiplication × should be set to zero.

12 Pending further methodological guidance to be developed by the Organization, the value of the multiplication × should be set to zero.

13 Pending further methodological guidance to be developed by the Organization, the value of should be set to zero.

Term Units Explanation
gCO2eq / g fuel Emissions associated with transport to long -term storage site
gCO2eq / g fuel Any emission associated with the process of storing (long-term: 100 years) the captured CO 2 (including fugitive emissions that may happen during long -term storage and/or the injection of CO2 into the storage)

site
gCO2eq / g fuel Any emission associated with the process of storing (long-term: 100 years) the captured CO 2 (including fugitive emissions that may happen during long -term storage and/or the injection of CO2 into the storage)
gCO2eq / g fuel Any additional emission related to the CCS
LCV MJ/g Lower Calorific Value is the amount of heat that would be released by the complete combustion of a specified fuel

5.3 In order to have LCA guidelines that will allow for their clear, robust and consistent application to any possible measure, the methodology allows to calculate two TtW values as follows:

.1 TtW GHG intensity value 1: calculated regardless of the carbon source, therefore the e c and eccu parameters should not be taken into account and the SFc and SFccu value should be always 0; and

.2 TtW GHG intensity value 2: calculated taking into account the carbon source for fuels of biogenic origins or made from captured carbon, therefore the ec and e ccu parameters should be taken into account and the S Fc and S Fccu values should be always 1.

5.4 The actual GHG intensity depends both on the properties of the fuel and on the

and e ccu parameters should be taken into account and the S Fc and S Fccu values should be always 1.

5.4 The actual GHG intensity depends both on the properties of the fuel and on the efficiency of the energy conversion. For CO2, the emission factors are based on the molar ratio of carbon to oxygen multiplied with the carbon mass of the fuel, assuming that all the carbon in the fuel is oxidized (stoichiometric combustion). The CH 4 and N 2O emissions factors are dependent on the combustion and/or conversion process in the energy converter.

5.5 For future use of, for example, fuel cells with a reforming unit, also electro -chemical reactions forming GHGs can be taken into account by this TtW methodology.

5.6 TtW default emission factors are provided in appendix 2 of these guidelines.

6 WELL-TO-WAKE (WtW)

6.1 The aim of the WtW methodology is to integrate WtT and TtW parts, to quantify the full life cycle emissions related to the production and use of a fuel.

6.2 The WtW GHG emission factor (gCO 2eq/MJLCV fuel or electricity) is calculated as follows:

Equation (3)

= +

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6.2 The WtW GHG emission factor (gCO 2eq/MJLCV fuel or electricity) is calculated as follows:

Equation (3)

 = +

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where:

Term	Units	Explanation
GHGWtW	gCO2eq/MJ(LCV)	Total well-to-wake GHG emissions per energy unit from the use of the fuel or electricity in a consumer on board the ship
GHGWtT	gCO2eq/MJ(LCV)	Total well-to-tank GHG upstream emissions per energy unit of the fuel provided to the ship
GHGTtW	gCO2eq/MJ(LCV)	Total tank-to-wake GHG downstream emissions per energy unit from the use of the fuel or electricity in a consumer on board the ship

Equation (4)

= + + +
+ 1 ((1 1100 (_ + C)) × (2 × 2 + 4 × 4 + 2 × 2) +(1100 (+ C) × ×) × ×)

Note: terms , and are pending further methodological guidance to be developed by the Organization.

Note: terms , and are pending further methodological guidance to be developed by the Organization. For more details refer to section 5.2.

6.3 For the purpose of calculating WtW, the TtW value 2 as calculated in accordance with paragraph 5.3.2 should be used.

7 SUSTAINABILITY

7.1 The sustainability of marine fuels should be assessed considering the following themes/aspects on a life cycle basis:

.1 greenhouse gases (GHG);
.2 carbon source;
.3 source of electricity/energy;
.4 carbon stock direct land use change (DLUC);
.5 carbon stock indirect land use change (ILUC);
.6 water;
.7 air;
.8 soil;
.9 waste and chemicals; and
.10 conservation.

Other social and economic sustainability themes/aspects may be considered at a later stage.

7.2 The principle/objective in conjunction with the associated metrics/indicators of each of the sustainability theme/aspect are specified below.

Table 1: Sustainability themes/aspects

Theme/aspect	Principle/Objective	Metric/Indicator
1. Greenhouse Gases (GHG)	Sustainable marine fuels generate lower GHG emissions than conventional marine fuels (energy-based weighted average of liquid petroleum products on 3 specific years of DCS data) on a life cycle basis.	1. GHG intensity in gCO2eq/MJ (GWP100); and GHG intensity in gCO2eq/MJ (GWP20) for comparative purposes.
2. Carbon source	Sustainable marine fuels do not increase GHG intensity from the use of fossil energy sources and the permanence of captured and stored carbon is ensured while also avoiding double counting across economic sectors.	1. Carbon source indicator, including its content (in %) and origin in feedstock used to produce final fuel product, i.e. Fossil, Biogenic, Captured Carbon (including direct air capture (DAC), point source fossil (PSF) and point source biogenic (PSB)), and Others (including mixture of

%) and origin in feedstock used to produce final fuel product, i.e. Fossil, Biogenic, Captured Carbon (including direct air capture (DAC), point source fossil (PSF) and point source biogenic (PSB)), and Others (including mixture of sources).
3. Source of electricity/energy
Sustainable marine fuels requiring significant electricity input during WtT phase and electricity delivered directly to ships are produced by using electricity/energy from renewable, nuclear or biogenic sources, which are additional to current or long-standing demand levels, or by using surplus electricity during off-peak hours.
1. The GHG intensity of electricity used in the production of marine fuels or delivered directly to ships (annual average, expressed in g CO_2eq/kWh based on total emissions and actual hours of production.
4. Carbon stock direct land use change (DLUC)

Sustainable marine fuels are not made from biomass obtained from land with high carbon stock; production of sustainable marine fuels minimizes emissions resulting from Direct Land Use Change.
1. Sustainable marine fuel feedstock does not include biomass obtained from land with high

obtained from land with high carbon stock; production of sustainable marine fuels minimizes emissions resulting from Direct Land Use Change.
1. Sustainable marine fuel feedstock does not include biomass obtained from land with high carbon stock (e.g. primary forests, wetlands, or peat lands referred to a specific cut-off date for conversion), or a sustainable land management plan and reporting schedule are in

14 Pending further guidance to be developed by the Organization.
Theme/aspect Principle/Objective Metric/Indicator
place to ensure that the biomass is obtained from activities or ecosystem services that do not negatively impact the soil carbon stock;
2. The production of sustainable marine fuels does not occur in lands converted from primary forest, forestland, grassland or legally protected land, taking (1 January 2008)14 as the cut-off date; and
3. Direct land-use change (DLUC) indicator, expressed in GHG (including CO_2, CH_4 and

converted from primary forest, forestland, grassland or legally protected land, taking (1 January 2008)14 as the cut-off date; and
3. Direct land-use change (DLUC) indicator, expressed in GHG (including CO2, CH4 and N2O emissions) intensity, i.e. mass of CO2 equivalent / MJ of production or yield of feedstock.
5. Carbon stock indirect land use change (ILUC)

Cultivation of feedstock of sustainable marine fuels minimizes inducing negative changes in the use or management of land which occurs outside the product system being assessed.
1. Indirect carbon stock risk associated with cultivation of feedstock for sustainable marine fuels (see para. 4.13).
6. Water

Production of sustainable marine fuels maintain or enhance water quality and availability.
1. Operational practices are in place to (1) maintain water quality; and (2) use water efficiently and to avoid the depletion of water resources (including surface water, renewable water and fossil/underground water) beyond replenishment capacities;
2. Respect of decision-making of local population on water management;
3. Water environment impact (weighted water consumption on water

surface water, renewable
water and
fossil/underground water)
beyond replenishment
capacities;
2. Respect of decision-
making of local population
on water management;
3. Water environment impact
(weighted water
consumption on water
scarcity);

Theme/aspect Principle/Objective Metric/Indicator
4. Water Use Indicator
expressed in m3/year per
MJ or production or yield of
feedstock;
5. Freshwater eutrophication
indicator, e.g. expressed in
kg of phosphorus
equivalent (Peq) and kg of
nitrogen equivalent (Neq)
released to fresh water/kg
of feedstock produced or
per MJ respectively; and
6. Marine eutrophication
indicator, e.g. expressed in
kg of phosphorus
equivalent (Peq) and kg of
nitrogen equivalent (Neq)
released to marine
water/kg of feedstock
produced or per MJ
respectively.
7. Air

Production of sustainable
marine fuels minimizes
negative impacts on air
quality.
1. The marine fuel is made
in a facility that fully
complies with all local,

	water/kg of feedstock produced or per MJ respectively.	
7. Air		
Production of sustainable marine fuels minimizes negative impacts on air quality.	1. The marine fuel is made in a facility that fully complies with all local, national and regional air pollution laws and regulations.	
8. Soil		
Production of sustainable marine fuels maintain or enhance soil health.	1. Agricultural and forestry best management practices for feedstock production or residue collection have been implemented to maintain or enhance soil health, such as physical, chemical and biological conditions; and 2. The marine fuel is made in a facility that fully complies with all local, national and regional laws and regulations about soil health.	
9. Waste and chemicals		
Production of sustainable marine fuels maintain or enhance responsible management of waste and use of chemicals.	1. Operational practices are implemented to ensure that waste arising from, and chemicals used in, production processes are minimized at storage, handling and disposal steps. Reuse or recycling of chemicals and waste is	

implemented to ensure
that waste arising from,
and chemicals used in,
production processes are
minimized at storage,
handling and disposal
steps. Reuse or recycling
of chemicals and waste is

Theme/aspect Principle/Objective Metric/Indicator
encouraged.
2. Procedures are in place
to minimize the use of
materials that are neither
recyclable nor
biodegradable;
3. Average (in tonnes) of
hazardous wastes
generated per MJ of fuel
produced; and
4. Average (in tonnes) of
specified industrial
chemicals consumed per
MJ of fuel produced.
10. Conservation

Production of sustainable
marine fuels maintain or
enhance biodiversity and
ecosystems, or conservation
services.

1. The marine fuel is not
made from feedstock
obtained from areas that
due to their biodiversity,
conservation value, or
ecosystem services, are
protected by the State
having jurisdiction over
the area. Evidence is
provided that the activity
does not interfere with the

obtained from areas that due to their biodiversity, conservation value, or ecosystem services, are protected by the State having jurisdiction over the area. Evidence is provided that the activity does not interfere with the protection purposes; and
2. Low invasive-risk feedstock is selected for cultivation and appropriate controls are adopted with the intention of preventing the uncontrolled spread of cultivated alien species and modified microorganisms.

8 FUEL LIFECYCLE LABEL (FLL)

8.1 The FLL is a technical tool to collect and convey the information relevant for the life cycle assessment of marine fuels and energy carriers (e.g. electricity for shore power) used for ship propulsion and power generation onboard in the context of these guidelines.

8.2 The FLL consists of five main parts, as illustrated below:

Part A-1	Part A-2	Part A-3	Part A-4	Part A-5
Fuel type (blend)	Fuel Pathway Code	Lower Calorific Value (LCV, MJ/g)	share in fuel blend	

8.2 The FLL consists of five main parts, as illustrated below:

Part A-1 Part A-2 Part A-3 Part A-4 Part A-5
Fuel type
(blend)
Fuel Pathway
Code
Lower Calorific
Value (LCV,
MJ/g)
share in fuel blend
(%MJ(LCV) / MJ(LCV))
WtT GHG emission
factor (GWP100,
gCO2eq/MJ(LCV))
+
Part B-1 (Part B-2)15
Emissions credits related to biogenic carbon
source (, in gCO2/g fuel based on
GWP100)
Emissions credits related to source of captured
carbon (, in gCO2/g fuel based on GWP100)
+
Part C-1 Part C-2 Part C-3
Value 1 (carbon source
NOT taken into account):
TtW GHG emission factor
(GWP100, gCO2eq/MJ(LCV))
Value 2 (carbon source
taken into account):
TtW GHG emission factor
(GWP100, gCO2eq/MJ(LCV))
Energy Converter
+
Part D Part E
WtW GHG emission factor
(GWP100, gCO2eq/MJ(LCV))
Note: Part D = Part A-5 + Part C-2
Sustainability (Certification)16

8.3 Different parties (fuel suppliers, owners/operators, Administration/RO, etc.) may use different parts of the FLL for different purposes along the fuel pathway. As such, each interested party may use those parts of the FLL as relevant to their activities and purposes

different parts of the FLL for different purposes along the fuel pathway. As such, each interested party may use those parts of the FLL as relevant to their activities and purposes rather than the complete, integrated document.

8.4 The five main parts of the FLL are explained below.

.1 Part A of the FLL indicates:

.1 fuel type (Part A-1);

.2 fuel pathway code (Part A-2);

.3 lower calorific value (Part A-3, in MJ/g); and

.4 WtT GHG emission factor (Part A-5, in gCO2eq/MJ(LCV) calculated on GWP100).

Part A-4 is only applicable when a fuel batch is supplied to the ship as a blend of fuels with different fuel pathway code (hereinafter referred to as the "fuel blend") and indicates the share of each blend component in the fuel

15 Pending further methodological guidance to be developed by the Organization (see section 5).

16 Pending further guidance to be developed by the Organization.

blend (in %MJ (LCV)/MJ(LCV)). If fuel blends are denoted on volume -basis, a re -calculation on energy basis based on the LCV values of the blend components is required;

For the fuel blend supplied to a ship, the information on fuel type for the mixture is presented under Part A -1 on top of its components, named by percentual order of composition in the fuel, e.g. X (70%), Y (20%), Z (10%). Part A-5, Part C-1, Part C-2 and Part D are the average value weighted on energy share (% MJ(LCV)) /MJ(LCV))) of each fuel component, while Part A-2 to A-4, Part B and Part E are kept blank. Each component of the fuel blend with a specific fuel pathways code is presented in a separate ro w below the row for the fuel blend;

.2 Part B of the FLL indicates the carbon credits related to the carbon source, including:

.1 ec (Part B-1, in gCO2 / g fuel calculated on GWP100); (and

.2 eccu (Part B-2, in gCO2/g fuel calculated on GWP100)),17

.2 Part B of the FLL indicates the carbon credits related to the carbon source, including:

.1 ec (Part B-1, in gCO2 / g fuel calculated on GWP100); (and

.2 eccu (Part B-2, in gCO2/g fuel calculated on GWP100)),17

as defined in section 5 of these guidelines;

.3 Part C of the FLL indicates the TtW GHG emission factor of the fuel type in conjunction with the energy converter(s) on board the ship (Part C -3). The TtW GHG emission factor of the fuel type is further categorized as:

.1 Value 1 where carbon source is not taken into account (Part C-1, in gCO2eq/MJ(LCV) calculated on GWP100); and

.2 Value 2 where carbon source is taken into account (Part C-2, in gCO2eq/MJ(LCV) calculated on GWP100),

as defined in section 5 of these guidelines;

.4 Part D of the FLL indicates the WtW GHG emission factor of the fuel type (in gCO2eq/MJ(LCV) calculated on GWP100), which is always the sum of Part A-5 and Part C-2; and

.5 Part E of the FLL indicates the sustainability performance of the fuel as per Section 7 of these guidelines.

PART III: DEFAULT EMISSION FACTORS AND ACTUAL VALUES

9 DEFAULT EMISSION FACTORS

A-5 and Part C-2; and
.5 Part E of the FLL indicates the sustainability performance of the fuel as per Section 7 of these guidelines.

PART III: DEFAULT EMISSION FACTORS AND ACTUAL VALUES

9 DEFAULT EMISSION FACTORS

9.1 The principles and the procedure described for the determination of default emission factors under this section 9 have been used for the establishment of default emission factors and should remain valid for the factors that will be established.

17 Pending further methodological guidance to be developed by the Organization. For more details on the eccu parameter and Part B-2 of the FLL, refer to sections 5.2 and 8.2, respectively.

9.2 WtT default emission factors should be calculated using representative and conservative assumptions, which encompass variable performance of feedstock -fuel pathways across world regions and States.

9.3 To establish a WtT default emission factor, at least three reference values from three

conservative assumptions, which encompass variable performance of feedstock -fuel pathways across world regions and States.

9.3 To establish a WtT default emission factor, at least three reference values from three different, representative, sources should be considered. Among the three (or more) values considered, the upper emission value should be selected as default, and the range of available emission factors should be provided for informative purposes. The reference values should be accompanied by the relevant technical and scientific information (see Template set out in appendix 4) and evaluated against the corresponding information as appropriate, including the agreement between the reference values.

9.4 Emissions related to carbon stock changes caused by direct land-use change (DLUC) (el) and emissions savings from soil carbon accumulation via improved agricultural management (esca) are considered as zero for the establishment of the initial default emission factors. Similarly, this is the case also for the parameters related to carbon capture and storage (ccs), which require further development.

management (esca) are considered as zero for the establishment of the initial default emission factors. Similarly, this is the case also for the parameters related to carbon capture and storage (ccs), which require further development.

9.5 TtW default emission factors, including slip factors per fuel type and per converter types, are set out in appendix 2 (for those fuels and converters for which such factors are available in resolution MEPC.364(79) on the 2022 Guidelines on the method of calculation of the attained energy efficiency design index (EEDI) for new ships and the Fourth IMO GHG Study 2020). Further TtW default emission factors (with the exception of C fCO2 provided in resolution MEPC.364(79)) may be established by following the same rules as for the WtT default emission factors described in paragraph 9.3. No default emission factors are provided for the use of on board CCS (eoccs), and the amount of captured carbon per unit of fuel mass should be specifically certified. T he parameters related to emission credits from the used captured CO2 as carbon stock to produce synthetic fuels (eccu) requires further development.

should be specifically certified. T he parameters related to emission credits from the used captured CO2 as carbon stock to produce synthetic fuels (eccu) requires further development.

9.6 As the definition of Cfug factors is considered a difficult parameter to be measured, Cfug factors should be established by the best existing knowledge and will be dealt with at a later stage. Until Cfug factors are defined, Cfug should be set as 0.

9.7 In case additional categories of energy converters (not listed in appendix 2) are proposed, the rules to establish TtW default emission factors as described in paragraph 9.5 above may be followed to ensure that these new converters (e.g. fuel cells) ma y also be associated with a default emission factor.

10 ACTUAL EMISSION FACTORS

10.1 The aim of actual emission factors is to allow demonstration of superior GHG performance compared to the default emission factors, subject to verification and certification by a third party.

10.2 WtT and TtW emission factors should be based on methodologies established in these guidelines. Actual values provide the WtW (WtT and TtW) GHG intensity for the specific

by a third party.

10.2 WtT and TtW emission factors should be based on methodologies established in these guidelines. Actual values provide the WtW (WtT and TtW) GHG intensity for the specific fuel over the life cycle (from fuel production to its use on board).

10.3 For the pathways contained in appendix 1, the description and the calculation method for providing WtT actual emission factors should be provided. In addition, for the pathways not contained in appendix 1, a detailed description of the pathway should be provided.

10.4 The use of actual WtT emission factors is not applicable to purely fossil pathways. However, for fuels which are produced from captured carbon of fossil origin and for fossil fuels where the technology of CCS/CCUS is applied, actual values are allowed. For the fossil component of a blended fuel, fossil fuel default emission factors should be used.

10.5 Actual TtW emission factors are allowed for all fuel pathways18 and provided in these guidelines.

PART IV: VERIFICATION AND CERTIFICATION

11 ELEMENTS SUBJECT TO VERIFICATION/CERTIFICATION

11.1 When used as evidence for performances, the FLL needs to be verified and certified by a third party, taking into account further guidance to be developed by the Organization.

11.2 The verification and certification of Part A, Part B, Part C, and Part E of the FLL may be carried out separately by different verification bodies. The verification and certification of Part D of the FLL needs to be based on the verified Part A, Part B and Part C.

11.3 For fuel types with a specific fuel pathway code and which will be consumed in a specified energy converter, the default emission factors for Part A -5, Part C-1, Part C-2 and Part D of the FLL are provided in appendix 2. As long as Part A-1 to Part A-4 and Part C-3 of

specified energy converter, the default emission factors for Part A -5, Part C-1, Part C-2 and Part D of the FLL are provided in appendix 2. As long as Part A-1 to Part A-4 and Part C-3 of the FLL have been duly verified, the default emission factors contained in these guidelines can be consequently applied without further verification.

11.4 In the case where lower emission factors are claimed compared to the default emission factors for Part A-5, Part C-1, Part C-2 and/or Part D, the actual emission factors can be used only after the verification and certification by a third party, taking int o account further guidance referred to in paragraph 11.1.

12 IDENTIFICATION OF CERTIFICATION SCHEMES/STANDARDS

12.1 The verification and certification of individual parts of the FLL will use relevant certification schemes/standards. Different parts of the FLL may be verified using different certification schemes/standards as applicable, while a specific part of the FLL may be addressed by multiple certification schemes/standards with similar scopes.

12.2 The certification schemes/standards used for the purposes specified in

certification schemes/standards as applicable, while a specific part of the FLL may be addressed by multiple certification schemes/standards with similar scopes.

12.2 The certification schemes/standards used for the purposes specified in paragraph 12.1 above should be recognized by the Committee, taking into account guidance to be developed by the Organization. The list of recognized certification schemes/standards should be publicly available and kept under review.

12.3 Proposals to recognize international certification schemes/standards should be submitted to the Committee for consideration, including an assessment of a set of predetermined criteria which will be further developed for this purpose.

12.4 The framework, criteria and procedures leading to the recognition of certification schemes should be implemented uniformly to guarantee the quality, reliability and robustness of the IMO framework as a whole and to ensure a level playing field among c ertification schemes.

18 Verification and certification methodologies would need further work to be established.
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)

schemes.

18 Verification and certification methodologies would need further work to be established.

PART V: REVIEW

13 CONTINUOUS REVIEW PROCESS

13.1 To ensure that new technological advances and scientific knowledge are taken into account, these guidelines should be kept under continuous technical review taking into account emerging and evolving technologies.

13.2 In particular, the following elements should be kept under review:

.1 WtT, TtW and WtW default emission factors as specified in appendix 2; and

.2 new proposed fuel pathways and the corresponding default emission factors in addition to those specified in appendix 1.

APPENDIX 1

FUEL LIST WITH FUEL PATHWAY CODES

Order Group Fuel type

APPENDIX 1

FUEL LIST WITH FUEL PATHWAY CODES

Order	Group	Fuel type	Feedstock structure	Conversion/Production process	Fuel Pathway Code
			Feedstock Type	Nature/Carbon Source	Process Type	Energy used in the process	
1	HFO (VLSFO)	Heavy Fuel Oil (ISO 8217 Grades RME, RMG and RMK, 0.10 < S 0.50%)	Crude Oil	Fossil	Standard refinery process	Grid mix electricity	HFO(VLSFO)_f_SR_gm
2	HFO (HSHFO)	Heavy Fuel Oil (ISO 8217 Grades RME, RMG and RMK exceeding 0.50% S)	Crude Oil	Fossil	Standard refinery process	Grid mix electricity	HFO(HSHFO)_f_SR_gm
3	LFO (ULSFO)	Light Fuel Oil (ISO 8217 Grades RMA, RMB and RMD maximum 0.10% S)	Crude Oil	Fossil	Standard refinery process	Grid mix electricity	LFO(ULSFO)_f_SR_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code

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Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
4 LFO
(VLSFO)
Light Fuel Oil
(ISO 8217
Grades
RMA, RMB
and RMD,
0.10 < S
0.50%)
Crude Oil Fossil Standard refinery
process
Grid mix electricity LFO(VLSFO)_f_SR_gm
5 Diesel/Ga
s oil
(ULSFO)
Marine
Diesel/Gas
Oil
(ISO 8217
Grades
DMX, DMA,
DMZ and
DMB
maximum
0.10 % S)
Crude Oil Fossil Standard refinery
process
Grid mix electricity MDO/MGO(ULSFO)_f_SR
_gm
6 Diesel/Ga
s oil
(VLSFO)
Marine
Diesel/Gas
Oil
(ISO 8217
Grades
DMX, DMA,
DMZ and
DMB, 0.10 <
S 0.50%)
Crude Oil Fossil Standard refinery
process
Grid mix electricity MDO/MGO(VLSFO)_f_SR
_gm

_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
7 Diesel/Ga
s oil
(ULSFO)
Bio co-
processed
marine fuel
(ISO 8217
Grades
DMX, DMA,
DMZ and
DMB
maximum
0.10 % S)
Crude Oil +
mixed
biomass
Fossil/Biogenic CoProcessing
(CP) in refinery
Grid mix electricity MDO/MGO(ULSFO)_f_b_
CP_gm
8 Diesel/Ga
s oil
(VLSFO)
Bio co-
processed
marine fuel
(ISO 8217
Grades
DMX, DMA,
DMZ and
DMB, 0.10 <
S 0.50%)
Crude Oil +
mixed
biomass
Fossil/Biogenic CoProcessing
(CP) in refinery
Grid mix electricity MDO/MGO(VLSFO)_f_b_
CP_gm
9 Diesel/Ga
s oil
(ULSFO)
Co-
processed
marine fuel

(ISO 8217
Grades
DMX, DMA,
DMZ and
DMB
maximum
0.10 % S)
Crude Oil +
recycled
carbon
Fossil/Recycled
carbon
CoProcessing
(CP) in refinery
Grid mix electricity MDO/MGO(ULSFO)_f_r_
CP_gm
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Grades
DMX, DMA,
DMZ and
DMB
maximum
0.10 % S)
Crude Oil +
recycled
carbon
Fossil/Recycled
carbon
CoProcessing
(CP) in refinery
Grid mix electricity MDO/MGO(ULSFO)_f_r_
CP_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
10 Diesel/Ga
s oil
(VLSFO)
Co-
processed
marine fuel
(ISO 8217
Grades
DMX, DMA,
DMZ and
DMB, 0.10 <
S 0.50%)
Crude Oil +
recycled
carbon
Fossil/Recycled
carbon
CoProcessing
(CP) in refinery
Grid mix electricity MDO/MGO(VLSFO)_f_r_
CP_gm
11 LPG19 Liquefied
Petroleum
Gas
(Propane)
Crude Oil Fossil Standard refinery
process and
liquefaction
Grid mix electricity LPG(Propane)_f_SR_gm
12 LPG Liquefied
Petroleum

Gas
(Propane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Fossil Steam
Methane
Reformation
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_fCO2_fH2

12 LPG Liquefied
Petroleum
Gas
(Propane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Fossil Steam
Methane
Reformation
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_fCO2_fH2
_FT_gm
13 LPG Liquefied
Petroleum
Gas
(Propane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture20

H2: from
Renewable
electricity
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_fCO2_rH2
_FT_gm

19 Regarding LPG, these guidelines consider the final product form the refineries to be always liquefied.

20 CO2: Fossil Point Source Carbon Capture includes captured CO2 stemming from fuel combustion and captured CO2 stemming from extraction of resources underground.

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
14 LPG Liquefied
Petroleum
Gas
(Propane)
CO2 + H2 CO2: Fossil Point

Order	Group Fuel type	Feedstock structure	Conversion/Production process	Fuel Pathway Code
Feedstock Type	Nature/Carbon Source	Process Type	Energy used in the process	
14	LPG Liquefied Petroleum Gas (Propane)	CO2 + H2	CO2: Fossil Point Source Carbon Capture	H2: Industrial by-product hydrogen Fischer-Tropsch Synthesis and liquefaction	Grid mix electricity	LPG(Propane)_fCO2_ibp H2_FT_gm
15	LPG Liquefied Petroleum Gas (Propane)	CO2 + H2	CO2: Direct Air Capture	H2: Fossil Steam Methane Reformation Fischer-Tropsch Synthesis and liquefaction	Grid mix electricity	LPG(Propane)_rCO2_fH2 _FT_gm
16	LPG Liquefied Petroleum Gas (Propane)	CO2 + H2	CO2: Direct Air Capture	H2: from Renewable electricity Fischer-Tropsch Synthesis and liquefaction	Grid mix electricity	LPG(Propane)_rCO2_rH2 _FT_gm
17	LPG Liquefied Petroleum Gas (Propane)	CO2 + H2	CO2: Direct Air Capture			

H2: Industrial by-
product hydrogen
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_rCO2_ibp
H2_FT_gm
18 LPG Liquefied
Petroleum
Gas
(Propane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Fossil Steam
Methane
Reformation

Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_rCO2_ibp
H2_FT_gm
18 LPG Liquefied
Petroleum
Gas
(Propane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Fossil Steam
Methane
Reformation
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_bCO2_fH2
_FT_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
19 LPG Liquefied
Petroleum
Gas
(Propane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: from
Renewable
electricity
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_bCO2_rH2
_FT_gm
20 LPG Liquefied
Petroleum
Gas
(Propane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-

product hydrogen
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_bCO2_ibp
H2_FT_gm
21 LPG Liquefied

Petroleum
Gas
(Propane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Propane)_bCO2_ibp
H2_FT_gm
21 LPG Liquefied
Petroleum
Gas (Butane)
Crude Oil Fossil Standard refinery
process and
liquefaction
Grid mix electricity LPG(Butane)_f_SR_gm
22 LPG Liquefied
Petroleum
Gas
(Butane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Fossil Steam
Methane
Reformation
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_fCO2_fH2_
FT_gm
23 LPG Liquefied
Petroleum
Gas
(Butane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: from
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_fCO2_rH2_
FT_gm

Order Group Fuel type

Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
Renewable
electricity

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
Renewable
electricity
24 LPG Liquefied
Petroleum
Gas
(Butane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Industrial by-
product hydrogen
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_fCO2_ibpH2
_FT_gm
25 LPG Liquefied
Petroleum
Gas
(Butane)
CO2 + H2 CO2: Direct Air
Capture

H2: Fossil Steam
Methane
Reformation
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_rCO2_fH2_
FT_gm
26 LPG Liquefied
Petroleum
Gas
(Butane)
CO2 + H2 CO2: Direct Air
Capture

H2: from
Renewable
electricity
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_rCO2_rH2_
FT_gm
27 LPG Liquefied
Petroleum
Gas
(Butane)

CO2 + H2 CO2: Direct Air
Capture

H2: Industrial by-
product hydrogen
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_rCO2_ibpH
2_FT_gm
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)

Petroleum
Gas
(Butane)
CO2 + H2 CO2: Direct Air
Capture

H2: Industrial by-
product hydrogen
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_rCO2_ibpH
2_FT_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
28 LPG Liquefied
Petroleum
Gas
(Butane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Fossil Steam
Methane
Reformation
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_bCO2_fH2_
FT_gm
29 LPG Liquefied
Petroleum
Gas
(Butane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: from
Renewable
electricity
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_bCO2_rH2_
FT_gm
30 LPG Liquefied

Petroleum
Gas
(Butane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Fischer-Tropsch

Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_bCO2_rH2_
FT_gm
30 LPG Liquefied
Petroleum
Gas
(Butane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Fischer-Tropsch
Synthesis and
liquefaction
Grid mix electricity LPG(Butane)_bCO2_ibpH
2_FT_gm
31 LNG Liquefied
Natural Gas
(Methane)
Natural Gas Fossil Standard LNG
production
including
liquefaction
Grid mix electricity LNG_f_SLP_gm
32 LNG Liquefied
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Thermochemical
gasification
followed by
methanation and
liquefaction
Grid mix electricity LNG_b_G_M_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
33 LNG Liquefied
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd

Gen.
feedstock
Biogenic Bio-derived LNG
via Anaerobic
Digestion,

Feedstock Type	Nature/Carbon Source	Process Type	Energy used in the process	
33	LNG Liquefied Natural Gas (Methane)	Mixed 1st, 2nd and 3rd Gen. feedstock	Biogenic Bio-derived LNG via Anaerobic Digestion, separation and liquefaction	Grid mix electricity LNG_b_AD_gm
34	LNG Liquefied Natural Gas (Methane)	Mixed 1st, 2nd and 3rd Gen. feedstock	Biogenic Bio-derived LNG via Anaerobic Digestion, separation with Point Source Carbon Capture (PSCC) and long-term storage and liquefaction	Grid mix electricity LNG_b_AD_CCS_gm
35	LNG Liquefied Natural Gas (Methane)	CO2 + H2	CO2: Fossil Point Source Carbon Capture H2: Fossil Steam Methane Reformation	Methanation and liquefaction	Grid mix electricity LNG_fCO2_fH2_M_gm
36	LNG Liquefied Natural Gas (Methane)	CO2 + H2	CO2: Fossil Point Source Carbon Capture H2: from Renewable electricity	Methanation and	

liquefaction
Grid mix electricity LNG_fCO2_rH2_M_gm
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
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Order Group Fuel type

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
37 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction
Grid mix electricity LNG_fCO2_ibpH2_M_gm
38 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Fossil Steam
Methane
Reformation
Methanation and
liquefaction
Grid mix electricity LNG_rCO2_fH2_M_gm
39 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: from
Renewable
electricity
Methanation and
liquefaction
Grid mix electricity LNG_rCO2_rH2_M_gm
40 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction

Grid mix electricity LNG_rCO2_ibpH2_M_gm
41 LNG Liquefied
Natural Gas
(Methane)

40 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction
Grid mix electricity LNG_rCO2_ibpH2_M_gm
41 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Fossil Steam
Methane
Reformation
Methanation and
liquefaction
Grid mix electricity LNG_bCO2_fH2_M_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
42 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: from
Renewable
electricity
Methanation and
liquefaction
Grid mix electricity LNG_bCO2_rH2_M_gm
43 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen

Methanation and
liquefaction
Grid mix electricity LNG_bCO2_ibpH2_M_gm
44 CNG Compressed

43 LNG Liquefied
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Methanation and
liquefaction
Grid mix electricity LNG_bCO2_ibpH2_M_gm
44 CNG Compressed
Natural Gas
(Methane)
Natural Gas Fossil Standard refinery
process and
compression
Grid mix electricity CNG_f_SR_gm
45 CNG Compressed
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Thermochemical
gasification
followed by
methanation and
compression
Grid mix electricity CNG_b_G_M_gm
46 CNG Compressed
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Bio-derived LNG
via Anaerobic
Digestion and
separation and
compression
Grid mix electricity CNG_b_AD_gm
47 CNG Compressed
Natural Gas
(Methane)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Bio-derived LNG
via Anaerobic
Digestion,
separation with
Grid mix electricity CNG_b_AD_CCS_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
Point Source
Carbon Capture
(PSCC) and long-
term storage and
compression
48 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Fossil Steam
Methane
Reformation
Methanation and
compression
Grid mix electricity CNG_fCO2_fH2_M_gm
49 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: from
Renewable
electricity
Methanation and
compression
Grid mix electricity CNG_fCO2_rH2_M_gm
50 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Industrial by-
product hydrogen
Methanation and
compression
Grid mix electricity CNG_fCO2_ibpH2_M_gm
51 CNG Compressed
Natural Gas
(Methane)

CO2 + H2 CO2: Direct Air Capture

H2: Fossil Steam Methanation and compression

Capture

H2: Industrial by-
product hydrogen
Methanation and
compression
Grid mix electricity CNG_fCO2_ibpH2_M_gm
51 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Fossil Steam
Methanation and
compression
Grid mix electricity CNG_rCO2_fH2_M_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
Methane
Reformation
52 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: from
Renewable
electricity
Methanation and
compression
Grid mix electricity CNG_rCO2_rH2_M_gm
53 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Direct Air
Capture

H2: Industrial by-
product hydrogen
Methanation and
compression
Grid mix electricity CNG_rCO2_ibpH2_M_gm
54 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic

Point Source
Carbon Capture

H2: Fossil Steam
Methane
Reformation

H2: Industrial by-
product hydrogen
Methanation and
compression
Grid mix electricity CNG_rCO2_ibpH2_M_gm
54 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Fossil Steam
Methane
Reformation
Methanation and
compression
Grid mix electricity CNG_bCO2_fH2_M_gm
55 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: from
Renewable
electricity
Methanation and
compression
Grid mix electricity CNG_bCO2_rH2_M_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
56 CNG Compressed
Natural Gas
(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Methanation and
compression
Grid mix electricity CNG_bCO2_ibpH2_M_gm
57 Ethane Ethane Natural Gas Fossil Standard refinery
process

(Methane)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Methanation and
compression
Grid mix electricity CNG_bCO2_ibpH2_M_gm
57 Ethane Ethane Natural Gas Fossil Standard refinery
process
Grid mix electricity Ethane_f_SR_gm
58 Vegetable
oil-based
fuel
Straight
Vegetable
Oil
1st Gen.
feedstock
Biogenic Extraction and
purification
Grid mix electricity SVO_b_EP _1stgen_gm
59 Vegetable
oil-based
fuel
Used oils
and fats
2nd Gen.
feedstock
Biogenic Extraction and
purification
Grid mix electricity UOF_b_EP _2ndgen_gm
60 Vegetable
oil-based
fuel
Algae oil 3rd Gen.
feedstock
Biogenic Extraction and
purification
Grid mix electricity AO_b_EP _3rdgen_gm
61 Diesel Diesel
(FAME)
1st Gen.
feedstock
Biogenic Transesterification Grid mix electricity FAME_b_TRE_1stgen_g
m_
62 Diesel Diesel
(FAME)
2nd Gen.
feedstock
Biogenic Transesterification Grid mix electricity FAME_b_TRE_2ndgen_g
m_
63 Diesel Diesel
(FAME)
3rd Gen.
feedstock

Biogenic Transesterification Grid mix electricity FAME_b_TRE_3rdgen_g
m_
64 Diesel Renewable
Diesel (Bio
FT-Diesel)
1st Gen.
feedstock
Biogenic Gasification and

m_
63 Diesel Diesel
(FAME)
3rd Gen.
feedstock
Biogenic Transesterification Grid mix electricity FAME_b_TRE_3rdgen_g
m_
64 Diesel Renewable
Diesel (Bio
FT-Diesel)
1st Gen.
feedstock
Biogenic Gasification and
Fischer-Tropsch
Synthesis
Grid mix electricity FT-
Diesel_b_G_FT_1stgen_g
m_
65 Diesel Renewable
Diesel (Bio
FT-Diesel)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Anaerobic
digestion and
methane
separation and
Grid mix electricity FT-Diesel_b_AD_FT_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
Fischer-Tropsch
Synthesis
66 Diesel Renewable
Diesel (Bio
FT-Diesel)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Anaerobic
digestion and
methane
separation and
Fischer-Tropsch
Synthesis with
Point Source

Carbon Capture (PSCC) and long-term storage
Grid mix electricity FT-Diesel_b_AD_FT_CCS_gm

Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Anaerobic
digestion and
methane
separation and
Fischer-Tropsch
Synthesis with
Point Source
Carbon Capture
(PSCC) and long-
term storage
Grid mix electricity FT-
Diesel_b_AD_FT_CCS_g
m
67 Diesel Renewable
Diesel (FT-
Diesel)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Fossil Steam
Methane
Reformation
Fischer-Tropsch
Synthesis
Grid mix electricity FT-
Diesel_fCO2_fH2_FT_gm
68 Diesel Renewable
Diesel (FT-
Diesel)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: from
Renewable
electricity
Fischer-Tropsch
Synthesis
Grid mix electricity FT-
Diesel_fCO2_rH2_FT_gm
69 Diesel Renewable
Diesel (FT-
Diesel)
CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

Fischer-Tropsch
Synthesis
Grid mix electricity FT-
Diesel_fCO2_ibpH2_FT_g
m

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon

Order Group	Fuel type	Feedstock structure	Conversion/Production process	Fuel Pathway Code	Feedstock Type	Nature/Carbon Source	Process Type	Energy used in the process
70	Diesel Renewable	Diesel (FT-Diesel)	CO2 + H2	CO2: Direct Air Capture	H2: Industrial by-product hydrogen			
71	Diesel Renewable	Diesel (FT-Diesel)	CO2 + H2	CO2: Direct Air Capture	H2: Fossil Steam Methane Reformation		Fischer-Tropsch Synthesis	Grid mix electricity
72	Diesel Renewable	Diesel (FT-Diesel)	CO2 + H2	CO2: Direct Air Capture	H2: from Renewable electricity		Fischer-Tropsch Synthesis	Grid mix electricity
73	Diesel Renewable	Diesel (FT-Diesel)	CO2 + H2	CO2: Biogenic Point Source Carbon Capture	H2: Industrial by-product hydrogen		Fischer-Tropsch Synthesis	Grid mix electricity

H2: Fossil Steam Methane Reformation Fischer-Tropsch Synthesis Grid mix electricity FT-Diesel_bCO2_fH2_FT_gm 74 Diesel Renewable Diesel (FT-Diesel) CO2 + H2 CO2: Biogenic Point Source Carbon Capture Fischer-Tropsch Synthesis

H2: Fossil Steam
Methane
Reformation
Fischer-Tropsch
Synthesis
Grid mix electricity FT-
Diesel_bCO2_fH2_FT_gm
74 Diesel Renewable
Diesel (FT-
Diesel)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture
Fischer-Tropsch
Synthesis
Grid mix electricity FT-
Diesel_bCO2_rH2_FT_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process

H2: from
Renewable
electricity
75 Diesel Renewable
Diesel (FT-
Diesel)
CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Fischer-Tropsch
Synthesis
Grid mix electricity FT-
Diesel_bCO2_ibpH2_FT_
gm
76 Diesel Renewable
Diesel (HVO)
1st Gen.
feedstock
Biogenic Hydrogenation Grid mix electricity HVO_b_HD_1stgen_gm_
77 Diesel Renewable
Diesel (HVO)
2nd Gen.
feedstock
Biogenic Hydrogenation Grid mix electricity HVO_b_HD_2ndgen_gm_
78 Diesel Renewable

Diesel (HVO)
3rd Gen.

feedstock
Biogenic Hydrogenation Grid mix electricity HVO_b_HD_1stgen_gm_
77 Diesel Renewable
Diesel (HVO)
2nd Gen.
feedstock
Biogenic Hydrogenation Grid mix electricity HVO_b_HD_2ndgen_gm_
78 Diesel Renewable
Diesel (HVO)
3rd Gen.
feedstock
Biogenic Hydrogenation Grid mix electricity HVO_b_HD_3rdgen_gm_
79 DME Dimethyl
Ether (DME)
1st Gen.
feedstock
Biogenic Gasification and
DME Synthesis
Grid mix electricity DME_b_G_DMES_1stgen
_gm_
80 DME Dimethyl
Ether (DME)
2nd Gen.
feedstock
Biogenic Gasification and
DME Synthesis
Grid mix electricity DME-b-G-
DMES_2ndgen_gm_
81 DME Dimethyl
Ether (DME)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Anaerobic
digestion and
methane
separation and
DME Synthesis
Grid mix electricity DME_b_AD_DMES_gm
82 DME Dimethyl
Ether (DME)
Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Anaerobic
digestion and
methane
separation and
Grid mix electricity DME_b_AD_DMES_CCS
_gm

Order Group Fuel type

_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
DME Synthesis
with Point Source
Carbon Capture
(PSCC) and long-
term storage
83 DME Dimethyl
Ether (DME)
Natural Gas Fossil Gasification and
DME Synthesis
Grid mix electricity DME_f_G_DMES_gm
84 Diesel Upgraded
Pyrolysis Oil
2nd Gen.
feedstock
Biogenic Pyrolysis, Fast
Pyrolysis, and/or
Catalytic Fast
Pyrolysis and
upgrading
Grid mix electricity UPO_b_UPO_2ndgen_gm
_
85 Diesel Hydrotherma
l Liquefaction
(HTL) Oil
2nd Gen.
feedstock
Biogenic Hydrothermal
liquefaction and
upgrading
Grid mix electricity HTL_b_HTL_2ndgen_gm_
86 Methanol Methanol Natural Gas Fossil Steam Methane
Reformation of
Natural Gas and
Methanol
Synthesis
Grid mix electricity MeOH_f_SMR_gm
87 Methanol Methanol Natural Gas Fossil Steam Methane
Reformation of
Natural Gas with
Carbon Capture &
Storage and

Reformation of
Natural Gas and
Methanol
Synthesis
Grid mix electricity MeOH_f_SMR_gm
87 Methanol Methanol Natural Gas Fossil Steam Methane
Reformation of
Natural Gas with
Carbon Capture &
Storage and
Methanol
Synthesis
Grid mix electricity MeOH_f_SMR_CCS_gm
88 Methanol Methanol Coal Fossil Gasification of
Coal and Methanol
Synthesis
Grid mix electricity MeOH_f_G_MS_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
89 Methanol Methanol Coal Fossil Gasification of
Coal with Carbon
Capture & Storage
and Methanol
Synthesis
Grid mix electricity MeOH_f_G_MS_CCS
_gm
90 Methanol Methanol 2nd and 3rd
Gen.
feedstock
Biogenic Gasification of
Biomass and
Methanol
Synthesis
Grid mix electricity MeOH_b_G_MS_gm
91 Methanol Methanol Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Reforming of
Renewable Natural
Gas (biomethane
from Anaerobic

feedstock
Biogenic Gasification of
Biomass and
Methanol
Synthesis
Grid mix electricity MeOH_b_G_MS_gm
91 Methanol Methanol Mixed 1st,
2nd and 3rd
Gen.
feedstock
Biogenic Reforming of
Renewable Natural
Gas (biomethane
from Anaerobic
Digestion) and
Methanol
Synthesis
Grid mix electricity MeOH_b_AD_MS_gm
92 Methanol Methanol CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Fossil Steam
Methane
Reformation
Methanol
Synthesis
Grid mix electricity MeOH_fCO2_fH2_MS_gm
93 Methanol Methanol CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: from
Renewable
electricity
Methanol
Synthesis
Grid mix electricity MeOH_fCO2_rH2_MS_g
m

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
94 Methanol Methanol CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Industrial by-
product hydrogen
Methanol

Synthesis

Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
94 Methanol Methanol CO2 + H2 CO2: Fossil Point
Source Carbon
Capture

H2: Industrial by-
product hydrogen
Methanol
Synthesis
Grid mix electricity MeOH_fCO2_ibpH2_MS_
gm
95 Methanol Methanol CO2 + H2 CO2: Direct Air
Capture

H2: Fossil Steam
Methane
Reformation
Methanol
Synthesis
Grid mix electricity MeOH_rCO2_fH2_MS_g
m
96 Methanol Methanol CO2 + H2 CO2: Direct Air
Capture

H2: from
Renewable
electricity
Methanol
Synthesis
Grid mix electricity MeOH_rCO2_rH2_MS_g
m
97 Methanol Methanol CO2 + H2 CO2: Direct Air
Capture

H2: Industrial by-
product hydrogen
Methanol
Synthesis
Grid mix electricity MeOH_rCO2_ibpH2_MS_
gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
98 Methanol Methanol CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
98 Methanol Methanol CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Fossil Steam
Methane
Reformation
Methanol
Synthesis
Grid mix electricity MeOH_bCO2_fH2_MS_g
m
99 Methanol Methanol CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: from
Renewable
electricity
Methanol
Synthesis
Grid mix electricity MeOH_bCO2_rH2_MS_g
m
100 Methanol Methanol CO2 + H2 CO2: Biogenic
Point Source
Carbon Capture

H2: Industrial by-
product hydrogen
Methanol
Synthesis
Grid mix electricity MeOH_bCO2_ibpH2_MS_
gm
101 Ethanol Ethanol 1st Gen.
feedstock
Biogenic Fermentation Grid mix electricity EtOH_b_FR_1stgen_gm_
102 Ethanol Ethanol 2nd Gen.
feedstock
Biogenic Pretreatment/hydr
olysis step and
Fermentation
Grid mix electricity EtOH_b_FR_2ndgen_gm_
103 Ethanol Ethanol 3rd Gen.
feedstock
Biogenic Fermentation Grid mix electricity EtOH_b_FR_3rdgen_gm_

olysis step and
Fermentation
Grid mix electricity EtOH_b_FR_2ndgen_gm_
103 Ethanol Ethanol 3rd Gen.
feedstock
Biogenic Fermentation Grid mix electricity EtOH_b_FR_3rdgen_gm_

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
104 Hydrogen Hydrogen Natural Gas Fossil Steam Methane
Reformation of
Natural Gas
Grid mix electricity H2_f_SMR_gm
105 Hydrogen Hydrogen Natural Gas Fossil Steam Methane
Reformation of
Natural Gas with
Carbon Capture
and long-term
storage
Grid mix electricity H2_f_SMR_CCS_gm
106 Hydrogen Hydrogen Natural Gas Fossil Methane Pyrolysis
into carbon and
hydrogen
Grid mix electricity H2_f_MPO_gm
107 Hydrogen Hydrogen Coal Fossil Gasification or
Carbonization of
Coal
Grid mix electricity H2_f_G_gm
108 Hydrogen Hydrogen Coal Fossil Gasification or
Carbonization of
Coal with Carbon
Capture and long-
term storage

107 Hydrogen Hydrogen Coal Fossil Gasification or
Carbonization of
Coal
Grid mix electricity H2_f_G_gm
108 Hydrogen Hydrogen Coal Fossil Gasification or
Carbonization of
Coal with Carbon
Capture and long-
term storage
Grid mix electricity H2_f_G_CCS _gm
109 Hydrogen Hydrogen 2nd Gen.
feedstock
Biogenic Gasification of
biomass and
Syngas separation
with Point Source
Carbon Capture
(PSCC) and long-
term storage
Grid mix electricity H2_b_G_SS_CCS_2ndge
n_gm_
110 Hydrogen Hydrogen Water +
Electricity
Renewable Dedicated
Photovoltaic
and/or Wind and/or
Renewable
electricity
LH2_EL_r_Liquefied

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
other Electrolysis
and liquefaction
111 Hydrogen Hydrogen Water +
Electricity
Fossil/Renewable Electrolysis and
liquefaction
Grid mix electricity LH2_EL_gm_Liquefied
112 Hydrogen Hydrogen Water +

other Electrolysis
and liquefaction
111 Hydrogen Hydrogen Water +
Electricity
Fossil/Renewable Electrolysis and
liquefaction
Grid mix electricity LH2_EL_gm_Liquefied
112 Hydrogen Hydrogen Water +
Electricity
Nuclear Thermochemical
Cycles or
Electrolysis and
liquefaction
Nuclear LH2_EL_n_Liquefied
113 Hydrogen Hydrogen Industrial by-
product hydrogen
Grid mix electricity LH2_ _ibp_gm _Liquefied
114 Ammonia Ammonia Natural Gas Fossil Methane Pyrolysis
into pure carbon
and hydrogen and
Haber Bosch
process
Grid mix electricity NH3_f_MPO_HB_gm
115 Ammonia Ammonia Natural Gas Fossil Steam Methane
Reformation of
Natural Gas and
Haber Bosch
process
Grid mix electricity NH3_f_SMR_HB_gm
116 Ammonia Ammonia Natural Gas Fossil Steam Methane
Reformation of
Natural Gas with
Point Source
Carbon Capture
(PSCC) and long-
term storage and
Grid mix electricity NH3_f_SMR_HB_CCS_g
m

Order Group Fuel type

m

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Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
Haber Bosch
process
117 Ammonia Ammonia Coal Fossil Gasification of
Coal and Haber
Bosch process
Grid mix electricity NH3_f_G_HB_gm
118 Ammonia Ammonia Coal Fossil Gasification of
Coal with Carbon
Capture and long-
term storage and
Haber Bosch
process
Grid mix electricity NH3_f_G_HB_CCS_gm
119 Ammonia Ammonia 2nd Gen.
feedstock
Biogenic Gasification Grid mix electricity NH3_b_G_2ndgen_gm_
120 Ammonia Ammonia N2 + H2 N2: separated with
renewable
electricity

H2: produced from
renewable
electricity
Haber Bosch
process
Grid mix electricity NH3_rN2_rH2_HB_gm
121 Ammonia Ammonia N2 + H2 N2: separated with
renewable
electricity

H2: Fossil Steam
Methane
Reformation
Haber Bosch
process
Grid mix electricity NH3_rN2_fH2_HB_gm

121 Ammonia Ammonia N2 + H2 N2: separated with
renewable
electricity

H2: Fossil Steam
Methane
Reformation
Haber Bosch
process
Grid mix electricity NH3_rN2_fH2_HB_gm

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
122 Ammonia Ammonia N2 + H2 N2: separated with
renewable
electricity

H2: Industrial by-
product hydrogen
Haber Bosch
process
Grid mix electricity NH3_rN2_ibpH2_HB_gm
123 Ammonia Ammonia N2 + H2 N2: separated with
grid mix electricity

H2: Fossil Steam
Methane
Reformation
Thermochemical
Cycles or
Electrolysis
Nuclear NH3_gmN2_fH2_EL_n
124 Ammonia Ammonia N2 + H2 N2: separated with
grid mix electricity

H2: produced from
renewable
electricity
Thermochemical
Cycles or
Electrolysis
Nuclear NH3_gmN2_rH2_EL_n
125 Ammonia Ammonia N2 + H2 N2: separated with
grid mix electricity

H2: Industrial by-
product hydrogen

grid mix electricity

H2: produced from
renewable
electricity
Thermochemical
Cycles or
Electrolysis
Nuclear NH3_gmN2_rH2_EL_n
125 Ammonia Ammonia N2 + H2 N2: separated with
grid mix electricity

H2: Industrial by-
product hydrogen
Thermochemical
Cycles or
Electrolysis
Nuclear NH3_gmN2_ibpH2_EL_n
126 Electricity Electricity Fossil/Renewable - Grid mix electricity Electricity_gm
127 Electricity Electricity Renewable Dedicated
Photovoltaic
and/or Wind and/or
other
Renewable
electricity
Electricity_renewable

Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
128 Wind
propulsion

APPENDIX 2

128 Wind
propulsion

APPENDIX 2

INITIAL DEFAULT EMISSION FACTORS PER FUEL PATHWAY CODE

Order Fuel type Fuel Pathway Code
WtT GHG
intensity
(gCO2eq/
MJ)
LCV
(MJ/g)
Energy
Convert
er
Cf
CO2
(gCO
2/g
fuel)
Cf CH4
(gCH4/g
fuel)
Cf N2O
(gN2O/g
fuel)
Cslip
/Cfug
(ma
ss
%)

ec
gC
O2eq
/g
fuel
TtW
GHG
intensity
(gCO2eq
/MJ)
NOTE
1
Heavy Fuel Oil
(ISO 8217 Grades

RME, RMG and
RMK, 0.10 < S
0.50%)
HFO(VLSFO)
_f_SR_gm 16.8 0.0402 ALL ICEs 3.114 0.00005 0.00018
Resolution
MEPC.364(79)
Fourth IMO
GHG study
2
Heavy Fuel Oil
(ISO 8217 Grades
RME, RMG and
RMK exceeding
0.50% S)
HFO(HSHFO)
_f_SR_gm 14.9
0.0402
ALL
ICEs 3.114 0.00005 0.00018
Resolution
MEPC.364(79)
Fourth IMO
GHG study
3
Light Fuel Oil (ISO
8217 Grades
RMA, RMB and
RMD maximum
0.10% S)
LFO(ULSFO)_
f_SR_gm
0.0412
ALL
ICEs 3.151 0.00005 0.00018
Resolution
MEPC.364(79)
Fourth IMO
GHG study
4
Light Fuel Oil (ISO
8217 Grades

Fourth IMO
GHG study
3
Light Fuel Oil (ISO
8217 Grades
RMA, RMB and
RMD maximum
0.10% S)
LFO(ULSFO)_
f_SR_gm
0.0412
ALL
ICEs 3.151 0.00005 0.00018
Resolution
MEPC.364(79)
Fourth IMO
GHG study
4
Light Fuel Oil (ISO
8217 Grades
RMA, RMB and
RMD, 0.10 < S
0.50%)
LFO(VLSFO)_
f_SR_gm
0.0412
ALL
ICEs 3.151 0.00005 0.00018
Resolution
MEPC.364(79)
Fourth IMO
GHG study

Order Fuel type Fuel Pathway Code
WtT GHG
intensity
(gCO2eq/
MJ)
LCV
(MJ/g)
Energy
Convert
er
Cf
CO2
(gCO
2/g
fuel)
Cf CH4
(gCH4/g
fuel)
Cf N2O
(gN2O/g

fuel)
Cslip
/Cfug
(ma
ss
%)

ec
gC
O2eq
/g
fuel
TtW
GHG
intensity
(gCO2eq
/MJ)
NOTE
5
Marine Diesel/Gas
Oil (ISO 8217
Grades DMX,
DMA, DMZ and
DMB maximum
0.10 % S)
MDO/MGO(U
LSFO)_f_SR_
gm
17.7 0.0427 ALL ICEs 3.206 0.00005 0.00018

Resolution
MEPC.364(79)
Fourth IMO
GHG study
6
Marine Diesel/Gas
Oil (ISO 8217
Grades DMX,
DMA, DMZ and
DMB, 0.10 < S
0.50%)
MDO/MGO(VL
SFO)_f_SR_g
m

0.10 % S)
MDO/MGO(U
LSFO)_f_SR_
gm
17.7 0.0427 ALL ICEs 3.206 0.00005 0.00018

Resolution
MEPC.364(79)
Fourth IMO
GHG study
6
Marine Diesel/Gas
Oil (ISO 8217
Grades DMX,
DMA, DMZ and
DMB, 0.10 < S
0.50%)
MDO/MGO(VL
SFO)_f_SR_g
m
0.0427 ALL ICEs 3.206 0.00005 0.00018

Resolution
MEPC.364(79)
Fourth IMO
GHG study
11
Liquefied
Petroleum Gas
(Propane)
LPG(Propane)
_f_SR_gm 0.0463 ALL ICEs 3.000 0.00005 0.00018
Resolution
MEPC.364(79)
Fourth IMO
GHG study
21
Liquefied
Petroleum Gas
(Butane)
LPG(Butane)_
f_SR_gm 0.0457 ALL ICEs 3.030 0.00005 0.00018
Resolution
MEPC.364(79)
Fourth IMO
GHG study

Order Fuel type Fuel Pathway Code
WtT GHG
intensity

(gCO2eq/
MJ)
LCV
(MJ/g)
Energy
Convert
er
Cf
CO2
(gCO
2/g
fuel)
Cf CH4
(gCH4/g
fuel)
Cf N2O
(gN2O/g
fuel)
Cslip
/Cfug
(mas
s %)

ec
gCO
2eq/g
fuel
TtW
GHG
intensity
(gCO2eq
/MJ)
NOTE
31 Liquefied Natural Gas (Methane) LNG_f_SLP_gm 0.0480
LNG
Otto
(dual

er
Cf
CO2
(gCO
2/g
fuel)
Cf CH4
(gCH4/g
fuel)
Cf N2O
(gN2O/g
fuel)
Cslip
/Cfug
(mas
s %)

ec
gCO
2eq/g
fuel
TtW
GHG
intensity
(gCO2eq
/MJ)
NOTE
31 Liquefied Natural Gas (Methane) LNG_f_SLP_gm 0.0480
LNG
Otto
(dual
fuel
medium
speed)
2.750 0 0.00011
3.5/-

Resolution
MEPC.364(79)
Fourth IMO
GHG study
LNG
Otto
(dual
fuel slow
speed)
1.7/-

LNG
Diesel
(dual
fuel slow
speed)
0.15/-

LBSI
(Lean-
Burn

Spark-
Ignited)
2.6/-

Steam
Turbines
and
boilers
0.01/-

Order Fuel type Fuel Pathway Code
WtT GHG
intensity
(gCO2eq/
MJ)
LCV
(MJ/g)
Energy
Convert
er
Cf
CO2
(gCO
2/g
fuel)
Cf CH4
(gCH4/g
fuel)
Cf N2O
(gN2O/g
fuel)
Cslip
/Cfug
(mas
s %)

ec
gCO
2eq/g
fuel
TtW
GHG
intensity
(gCO2eq
/MJ)
NOTE
33 Liquefied Natural Gas (Methane) LNG_b_AD_gm
LNG
Otto
(dual
fuel
medium
speed)
2.750

LNG Otto (dual fuel slow speed)
LNG Diesel (dual fuel slow speed)
LBSI (Lean-Burn Spark-

TtW
GHG
intensity
(gCO2eq
/MJ)
NOTE
33 Liquefied Natural Gas (Methane) LNG_b_AD_gm
LNG
Otto
(dual
fuel
medium
speed)
2.750

LNG
Otto
(dual
fuel slow
speed)
LNG
Diesel
(dual
fuel slow
speed)
LBSI
(Lean-
Burn
Spark-
Ignited)
Steam
Turbines
and
boilers
62 Diesel (FAME) FAME_b_TRE_gm_2ndgen 20.8 0.0372 ALL ICEs
77 Renewable Diesel (HVO) HVO_b_HD_gm_1stgen 14.9 0.044 ALL ICEs

Order Fuel type Fuel Pathway Code
WtT GHG
intensity
(gCO2eq/
MJ)
LCV
(MJ/g)
Energy
Convert
er
Cf
CO2
(gCO
2/g
fuel)

| | | |

121 Ammonia NH3_rN2_fH2_HB_gm 0.0186 ALL ICEs 0 Fuel cell

APPENDIX 3

ABBREVIATIONS AND GLOSSARY

Abbreviations

AR IPCC Assessment Report
BDN Bunkering Delivery Note
Cf Emission conversion factors C fCO2/CH4/N2O (g GHG (CO 2/CH4/N2O)/g fuel) for emissions of the combustion and/or oxidation process, including the fuel with relevant GWP effect resulting from the combustion energy conversion
CH4 Methane
CO2 Carbon dioxide
CO2eq Carbon dioxide equivalent
CCS Carbon Capture and Storage
CCU Carbon Capture and Utilization
DAC Direct Air Capture
DCS IMO ship fuel oil consumption Data Collection System
DLUC Direct Land Use Change
FLL Fuel Lifecycle Label
GHG Greenhouse gas
GWP Global Warming Potential
ILUC Indirect Land Use Change
IPCC Intergovernmental Panel on Climate Change
LCA Life Cycle Assessment
LCV Lower Calorific Value (MJ/g fuel)
NMVOC Non-Methane Volatile Organic Compounds
N2O Nitrous oxide

ILUC Indirect Land Use Change
IPCC Intergovernmental Panel on Climate Change
LCA Life Cycle Assessment
LCV Lower Calorific Value (MJ/g fuel)
NMVOC Non-Methane Volatile Organic Compounds
N2O Nitrous oxide
NTC NOx Technical Code
RFNBO Renewable Fuels of Non-Biological Origin
SLCF Short-Lived Climate Forcers
TtW Tank-to-Wake
WtT Well-to-Tank
WtW Well-to-Wake
VOC Volatile Organic Compounds

Glossary

Co-product an outcome of a production process, which has a relevant economic value and elastic supply (intended as the existence of a clear evidence of the causal link between feedstock market value and the quantity of feedstock that can be produced).

Biomass Biomass is renewable organic material that comes from plants and animals.

Renewables any form of energy from solar, geophysical or biological sources that is replenished by natural processes at a rate that equals or exceeds its rate of use. Renewables are obtained from the continuing or repetitive flows of energy occurring in the natural environment and includes low-carbon technologies such as solar energy, hydropower, wind, tide and waves and ocean

from the continuing or repetitive flows of energy occurring in the natural environment and includes low-carbon technologies such as solar energy, hydropower, wind, tide and waves and ocean thermal energy, as well as renewable fuels such as biomass.

Global Warming Potential Global Warming Potential indicates the potential of a greenhouse gas to trap extra heat in the atmosphere over time in relation to carbon dioxide. The enhanced heat trapping in the atmosphere (i.e. the "greenhouse effect") is caused by the absorption of infrared radiation by a given gas. The GWP also depends on the atmospheric lifetime of a gas, and the
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
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time-horizon considered (for example, GWP20 is based on the energy absorbed over 20 years, whereas GWP100 is based on the energy absorbed over 100 years. Each greenhouse gas has a specific global warming potential which is used to calculate the CO2-equivalent (CO2eq).

whereas GWP100 is based on the energy absorbed over 100 years. Each greenhouse gas has a specific global warming potential which is used to calculate the CO2-equivalent (CO2eq).

Land Use Change - Production of bio-based fuels leads to land use change (LUC). LUC can be classified as direct LUC (DLUC) and indirect LUC (ILUC).

Life Cycle Assessment (LCA) framework Life Cycle Assessment determines the potential environmental impacts of products, processes or services from cradle to grave, e.g. from acquisition/extraction of raw materials through to processing, transport, use and disposal.

System boundaries The system boundary determines which entities (unit processes) are inside the system and which are outside. It essentially determines which life cycle/supply chain stages and processes are included in the assessment and need to be in accordance with the goal and scope of the study.

System expansion ISO 14040 recommends the use of system expansion whenever possible. System expansion is part of the consequential LCA method that seeks to capture change in environmental impact as a consequence of a certain activity.

System expansion ISO 14040 recommends the use of system expansion whenever possible. System expansion is part of the consequential LCA method that seeks to capture change in environmental impact as a consequence of a certain activity.

Well-to-Wake - WtW studies estimate the energy requirements and the resulting greenhouse gas (GHG) emissions in the production of a fuel and its use in a ship, based on the broader Life Cycle Assessment (LCA) methodology. The term 'Well' is used for fuels from all sou rces, because although the term is most applicable to conventional crude oil resources, it is widely used and understood.

RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
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APPENDIX 4

TEMPLATE FOR WELL-TO-TANK DEFAULT EMISSION FACTOR SUBMISSION

1 Explanatory remarks on the general scope of the template: This template aims at collecting and presenting in a clear and structured manner the input data used to calculate a "default

1 Explanatory remarks on the general scope of the template: This template aims at collecting and presenting in a clear and structured manner the input data used to calculate a "default emission factor" for a specific "feedstock -to-fuel" pathway. A "default emission factor" aims as representing the quantitative results of a high-level assessment about the carbon intensity (gCO2eq/MJ) of a feedstock-to-fuel value chain. The default emission factor is not meant to represent the best available way to produce a fuel, rather a value potentially describing a feedstock production then converted in a standard plant, located in a generic region. A default emission factor does not have to capture process improvement, with respect to current production, nor innovative technologies. The goal of default emission factor is, at least, twofold:

.1 allow for a carbon intensity comparison among different technologies;

.2 allow for operators to demonstrate lower core life cycle emissions compared to the default core life cycle, thought a certification process.

2 Operators (e.g. fuel producers) can ask for being certified, in order to prove better

.2 allow for operators to demonstrate lower core life cycle emissions compared to the default core life cycle, thought a certification process.

2 Operators (e.g. fuel producers) can ask for being certified, in order to prove better performances than the default emission factor (that cannot therefore be the representation of the best available technology), obtain a certified "actual value". Actual values may also be used when the fuel producer has defined a new pathway that does not have a default core life cycle emission factor.

3 This template allows presenting the minimum set of data required for the calculation of default core LCA emission factors, ensuring quality in terms of data relevance, adequacy, quality, transparency and accessibility.

PATHWAY DESCRIPTION

4 This section should clearly present the pathway modelled, with the aim for providing at least information on: the type of feedstock used, a description of the technology used for converting such feedstock in the final fuel, and any other relevant information, consistently with the system boundary of the LCA guidelines.

least information on: the type of feedstock used, a description of the technology used for converting such feedstock in the final fuel, and any other relevant information, consistently with the system boundary of the LCA guidelines.

5 Explanatory remarks on the pathway description: The default emission factors are based on the WtT methodology, aiming at evaluating the amount of GHG emissions for the fuel production and distribution. The production steps to be included in the WtT are:

Figure 2 Generic well-to-tank supply chain

The system boundaries defined for describing a specific feedstock-to-fuel pathway shall be in line with the definitions contained in the guidelines.

Additional details and relevant information may be added in appendices, such as location, production capacity, age of production facility or facilities.

INPUT DESCRIPTION

6 This section should clearly present the input used for the modelling exercise.

INPUT DESCRIPTION

6 This section should clearly present the input used for the modelling exercise.

7 Source of the data and of the used model should be reported.

8 Explanatory remarks on the input description: In order to provide guidance to fill the template, please see below some tables aiming at presenting the data that should be reported, per pathway (example based on a lipid feedstock production and conversion). As, in practical terms, the tables are "pathway specific", please adapt when needed.

Table 1: efecu inputs and outputs for XXX feedstock

	XXXX, per dry kg	Data source/Model used
XXX feedstock Agricultural Inputs	Total N (g)	zzz et al. 2010
efecu	P2O5 (g)	ecoinvent
	K2O (g)	GREET
Diesel (MJ)		
per kg XXXX oil Oil Extraction Inputs		
	Values	Data source/ Model used
Feedstock (g, dry)		zzz et al. 2010
NG (MJ)		ecoinvent
N-Hexane (MJ)		GREET
Electricity (MJ)		
Electricity (MJ)		
Co-product, zzz(g)		
Co-product, zzz (g)		

Feedstock
(g, dry)
zzz et al. 2010
NG (MJ) ecoinvent
N-Hexane
(MJ)
GREET
Electricity
(MJ)

Electricity
(MJ)

Co-
product,
zzz(g)

Co-
product,
zzz (g)

Oil Extraction
Outputs
Co-
product,
zzz (g)

Table 2: ep Inputs and outputs for XXXX conversion process Explanatory remark:
including all the needed steps to pre-treat the feedstock in order to be able
to convert it into the fuel, via the selected conversion process.

per MJ fuel
Values Data source/model used
Inputs
Feedstock (g oil) zzz et al. 2010
NG (MJ) ecoinvent
H2 (MJ) * GREET
Electricity (MJ)
Explanatory remark:
placeholder for key
material inputs (e.g.
chemicals, etc.)

Outputs
Co-product, propane
mix (MJ) **

Co-product, naphtha (MJ) **
Co-product, xxxx (MJ)

Electricity (MJ)
Explanatory remark:
placeholder for key
material inputs (e.g.
chemicals, etc.)

Outputs
Co-product, propane
mix (MJ) **
Co-product, naphtha
(MJ) **
Co-product, xxxx (MJ)

*H2 derived from NG steam reforming, included in NG input; ** Inputs after allocation

Table 3: Inputs for regional electricity generation mixes

	US (%)1	EU (%)2	India3 (%)	Xxx (%)
Residual oil				
Natural gas				
Coal				
Nuclear power				
Biomass				
Hydroelectric				
Geothermal				
Wind				
Solar PV				
Others				

1 GREET 20xx, 2EEA, 20xx (EU electricity mix 20xx), 3International Energy Agency 20xx.

Table 4: etd Inputs for transportation of feedstock and fuels Explanatory remark: in filling

Table 4: etd Inputs for transportation of feedstock and fuels Explanatory remark: in filling the table, please add the fuel used - In the "Data source/model used" please specify the type of fuel, the specific efficiency and energy converter, if available.

Feedstock Transportation	Data source/model used
etd Inputs for Transport and Distribution	
Distance (km) xxx; xxx Mode Heavy-duty truck; Train; Ship Share (%) yy; yy; yyy Fuel Transportation Distance (km) xxx; xxxx; xx Mode Barge; Rail; Heavy-duty truck Share (%) y; yy; yy Fuel Distribution Distance (km) xx Mode Heavy-duty truck Share (%)	
Any other Transportation and Distribution	

MAIN RESULTS

9 This section should present the results of the modelled pathway.

Table 5: Fuel identification

Fuel Pathway Code	LCV (MJ/g)	Density (kg/m3)	CfCO2

Transportation
and
Distribution

MAIN RESULTS

9 This section should present the results of the modelled pathway.

Table 5: Fuel identification

Fuel Pathway Code LCV (MJ/g) Density (kg/m3) CfCO2

Table 6: Proposed default emission factors for XXX-converted in a YYYY pathway

Region
efecu
Feedstock
cultivation/extraction
etd
Feedstock
transportation
ep
Fuel
production
(Sum of the
terms)
Proposed
WtT GHG
intensity
(gCO2eq/MJ)
emission
factors
ZZZZ
AAAA
BBBB

APPENDIX

10 Brief description of the pathway

11 Brief description of the technology
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REFERENCES

12 REF (APA format)

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RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES)